```
class GPModel(gpytorch.models.ExactGP, botorch.models.gpytorch.GPyTorchModel):  ❶
    num_outputs = 1  ❶

    def __init__(self, train_x, train_y, likelihood):
        super().__init__(train_x, train_y, likelihood)
        self.mean_module = gpytorch.means.ConstantMean()
        self.covar_module = gpytorch.kernels.ScaleKernel(
            gpytorch.kernels.RBFKernel()
        )

    def forward(self, x):
        mean_x = self.mean_module(x)
        covar_x = self.covar_module(x)
        return gpytorch.distributions.MultivariateNormal(mean_x, covar_x)
```

❶ Modifications for BoTorch integration.

Aside from these, everything else in our GP implementation remains the same. We now write a helper function that trains the GP on our training data:

```
def fit_gp_model(train_x, train_y, num_train_iters=500):
    noise = 1e-4  ❶

    likelihood = gpytorch.likelihoods.GaussianLikelihood()  ❶
    model = GPModel(train_x, train_y, likelihood)  ❶
    model.likelihood.noise = noise  ❶

    optimizer = torch.optim.Adam(model.parameters(), lr=0.01)  ❷
    mll = gpytorch.mlls.ExactMarginalLogLikelihood(likelihood, model)  ❷

    model.train()  ❷
    likelihood.train()  ❷

    for i in tqdm(range(num_train_iters)):  ❷
        optimizer.zero_grad()  ❷

        output = model(train_x)  ❷
        loss = -mll(output, train_y)  ❷

        loss.backward()  ❷
        optimizer.step()  ❷

    model.eval()  ❷
    likelihood.eval()  ❷

    return model, likelihood
```

❶ Declare the GP.

❷ Train the GP using gradient descent.

> **NOTE** **We have used all the code above in previous chapters. If you are having trouble understanding a piece of code, refer to Section 3.3.2 for more details.**

This covers Step 1 of Figure 4.2. For now, we are skipping Step 2, which is about implementing BayesOpt policies, and saving it for the next section and future chapters. The next component to

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